MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU–BVOC (HONOURS) Regular EXAMINATION – November 2025
FIRST SEMESTER
Skill Development Components (SDC) – MG1SDCAFP101 – BAKERY AND CONFECTIONARY
(2025 Admission Onwards)

Duration: 1 Hour 30 Minutes

Maximum Marks: 70

PART A

1. Write the differences between sulphitation and carbonation.

Ans :

Sulphitation:
• Involves adding sulphur dioxide (SO_2) to cane juice.
• Prevents browning and acts as a preservative.

Carbonation:
• Involves addition of lime ($Ca(OH)_2$) and CO_2 to remove impurities.
• Used for clarification of juice.

2. What are confectionerys?

Ans : Confectionery refers to sweet food items made mainly of sugar, such as candies, chocolates, toffees, fudge, jellies, lozenges, and chewing gums.

3. What is alkalisation of cocoa beans?

Ans : Alkalisation is the process of treating cocoa nibs or cocoa liquor with an alkali (e.g., potassium carbonate) to improve flavor, color, and dispersibility.

4. What are unshortened cakes?

Ans :

- Also called foam cakes.
- Do not contain fat.
- Egg whites are beaten to trap air.
- Leavened by air and steam during baking.
- Texture is light, fluffy, and spongy.
- Very little or no oil used.
- Examples: angel food cake, sponge cake.

What is gur?

Ans : Gur, also called jaggery, is a natural sweetener made from sugarcane juice. It is brown or golden in color and is used in many Indian sweets. Gur is healthier than white sugar and is full of natural minerals. Sources of Gur: Mainly from sugarcane Also made from palmyra, date palm, and coconut sap . Color - Golden yellow to brown

5. What is beet sugar?

Ans : Beet sugar is the sugar made from the root of the sugar beet plant. The beet root contains a high amount of sucrose (sugar). It is an alternative to cane sugar and is mostly vegan.

6. What do you mean by honeycomb candy?

Ans : A light, porous candy made by adding baking soda to hot sugar syrup, which releases CO_2 and forms a honeycomb texture

7. What is double–boiling method?

Ans :

Double Boiling Method - Heat a little water in the bottom pan and place a bowl with chocolate on top. • The chocolate melts slowly with the steam heat. • Stir occasionally to make it smooth. • Be careful — even a drop of water can spoil the chocolate. • This method helps avoid overheating.

8. What are the two types of biscuits?

Ans :

• **Hard dough biscuits** (Marie, Glucose)
• **Soft dough biscuits** (Cookies, Shortbread)

9. What is knock back?

Ans :

- Knock back means punching or pressing the dough during the fermentation period.
- It helps to release some of the gas formed inside the dough.
- This process improves the gas-retaining capacity of the dough.
- It also helps to mix the dough evenly and make it smooth.

10. What is refined sugar?

Ans :

Refined sugar is made from raw sugar (from sugarcane or sugar beet) that undergoes a refining process to remove impurities like molasses. It is also known as table sugar or white sugar. Commonly used in cooking, baking, and processed foods.

11. What are the ingredients used in lollipop?

Ans : Sugar, glucose syrup, water, flavoring, coloring, acidulants (citric acid), and sometimes stabilizers.

Part B

13. Explain the difference between dark and wafer coated chocolate.

Ans :

Dark Chocolate:
• Contains cocoa mass, cocoa butter, sugar.
• No milk solids.
• Higher cocoa percentage and intense flavor.

Wafer-coated Chocolate:
• Chocolate coating over wafer layers.
• Contains milk solids, vegetable fats, emulsifiers.
• Used for confectionery covering rather than pure eating chocolate.

14. Write a note on fermentation during bread making.

Ans :

- Fermentation is the process where yeast acts on sugars in the dough.
- It is an anaerobic process that takes place without oxygen.
- It starts immediately after mixing and continues during inter proofing and final proofing.
- The main purpose of fermentation is to produce carbon dioxide gas which makes the dough rise.
- It also helps to develop flavour and aroma in the bread.
- Fermentation improves the dough texture and makes it soft.
- Ideal temperature for fermentation is between 24°C and 30°C.
- Ideal humidity is about 70% to 75%.
- These conditions are best for yeast activity and proper fermentation.
- Usually 2% baker's yeast (based on flour weight) is used.
- Too little yeast causes slow fermentation.
- Too much yeast gives sour Flavors and poor texture.
- During fermentation, glucose (sugar) is converted by yeast into carbon dioxide and alcohol.
- Carbon dioxide makes the dough rise, and alcohol evaporates during baking.

- Well-fermented dough becomes light, soft, and airy.
- Bread made from properly fermented dough has good volume, taste, and texture.

15. Explain the different tempering methods.

1. Direct Method In this method, chocolate is melted directly in a microwave oven. • Heat the chocolate at 50% power and check every 20 seconds. • When most of it melts and a few small lumps remain, take it out. • Stir well — the remaining lumps will melt with the heat. • This method is quick and easy, but if heated too long, the chocolate can overheat and burn.

2. Double Boiling Method • Heat a little water in the bottom pan and place a bowl with chocolate on top. • The chocolate melts slowly with the steam heat. • Stir occasionally to make it smooth. • Be careful — even a drop of water can spoil the chocolate. • This method helps avoid overheating.

3. Heating Pad Method • Place the bowl of chocolate on a heating pad set to low heat. • The chocolate melts slowly and evenly. • This method makes it easy to control the temperature and avoid overheating. • It is useful for keeping chocolate melted and warm while working with it.

4. Seeding Method • This is the most common professional method. • Melt 2/3 of the chocolate and then add 1/3 chopped solid chocolate (called seeds). • Stir until the added pieces melt completely. • The solid pieces help cool and stabilize the melted chocolate, giving it a smooth, shiny finish.

16. What is sugarcane? Write down its nutritive value.

Sugarcane

Sugarcane is the **chief source of sugar**. It contains **12–15% sugar** (mainly sucrose). In India, about **80–90% of sugarcane** is used for the **manufacture of sugar**, **gur**, **khandsari**, and other products.
It is also used for making **beverages**, **chewing cane**, and **various sugar products**.

Nutritive Value of Sugarcane Juice

Nutrient	Amount
Energy	40–60 kcal
Carbohydrates	10–15 g
Sugars (Sucrose, Glucose, Fructose)	9–13 g
Protein	0.1 g
Fat	0 g

Nutrient	Amount
Fiber	0 g
Calcium	10–15 mg
Potassium	150–200 mg
Iron	0.3 mg
Magnesium	10–12 mg

- Sugarcane contains **sucrose, glucose, and fructose**.
- Used for **white sugar**, **gur (jaggery)**, **khandsari**, and **molasses**.
- **5–10%** of sugarcane is used for **chewing**.
- It provides **quick energy** due to high sugar content.
- Good source of **potassium**, **calcium**, and **iron**.

17. Explain the method of processing fudge.

Fudge

Fudge is a type of Western confectionery that is **soft, sweet, creamy, and rich**. It is made by cooking sugar, milk, butter, and flavouring ingredients.

- Usually soft and smooth in texture.
- Principle of making fudge is similar to that of making **fondant**.
- Sold in different flavours such as chocolate, vanilla, caramel, and nuts or fruits may be added.

Ingredients

- Sugar – 200 g
- Milk – 120 g
- Butter – 15 g
- Chocolate – 20 g

Method

1. Melt chocolate and butter together.
2. Add sugar and mix well.
3. Add milk and heat until the sugar dissolves completely.
4. Cook the syrup to **soft ball stage (112°C)**.
5. Allow it to cool to about **70°C**.
6. Transfer to a greased pan, let it set, and cut into pieces.

18. How cocoa powder is produced?

Steps in Cocoa Powder Production

Harvesting

Ripe cocoa pods are cut from the trees using knives or machetes.

Pod Opening

The pods are split open and the beans with pulp are removed.

Fermentation

Beans are fermented for **5–7 days** to develop flavour, reduce bitterness, and remove pulp.

Drying

Fermented beans are dried in the sun or dryers until moisture reaches **6–7%**.

Breaking and Winnowing

Dried beans are cracked (breaking) and shells are removed to get **cocoa nibs**.

Alkalisation (Dutching)

Nibs are treated with alkali (e.g., potassium carbonate) to improve colour, flavour, and dispersibility.

Roasting

Alkalised nibs are roasted at **120–150°C** to enhance flavour and reduce moisture.

Nib Grinding

Roasted nibs are ground into a thick paste called **cocoa liquor** (or cocoa mass).

Liquor Treatment (Pressing)

Cocoa liquor is pressed to separate **cocoa butter**.
The solid cake left behind is used for cocoa powder.

Pulverisation

The cocoa cake is crushed and ground into **fine cocoa powder**.

Packing

The cocoa powder is sieved and packed in **airtight, moisture-proof containers or bags** to prevent spoilage.

Storage

Packed powder is stored in **cool, dry conditions**, away from light and strong odours to maintain quality.

Transport

Finished cocoa powder is transported in sealed containers or cartons to protect it from moisture, heat, and contamination.

Part C

19. Describe the role of essential ingredients in bread making.

ROLE OF INGREDIENTS

1. Wheat Flour

• Major ingredient in bread making.
• Obtained through the milling process, where the bran and germ parts of the wheat grain are removed.

Main components of flour:
o Moisture – 14%
o Starch – 70%
o Protein – 11%
o Ash – 1.5%
o Sugar – 1%
o Fat – 1%

Characteristics of Good Quality Flour :

Appearance
• Flour should be creamy white and smooth.
• It should not contain bran pieces (brown flakes from wheat's outer layer).

Strength
• Flour should be strong so the dough keeps its shape.
• This strength comes from good gluten, which helps the bread rise well.

Water Absorption
• Good flour absorbs more water and forms a soft, stretchy dough.
• It needs less mixing to form properly.
• This makes the bread light and fluffy after baking.

2. Water

• Main role: Water is an essential ingredient in bread making.

1. Helps form gluten – Water mixes with flour proteins (gliadin and glutenin) to make elastic gluten, which gives the dough structure.
2. Controls dough temperature – Keeps the dough at the right temperature for yeast to work and ferment properly.
3. Dissolves ingredients – Helps mix and spread salt, sugar, fat, milk powder, and yeast evenly in the dough.
4. Softens starch – Makes the starch in flour soft and easy to digest after baking.
5. Activates enzymes – Helps enzymes work, which supports fermentation and dough development.
6. Improves texture – Makes bread soft, smooth, and pleasant to eat.

3. Salt

• Adds taste and makes the bread more flavourful.
• Gives the bread its typical pleasant flavour.
• Controls yeast activity – slows it down so the dough doesn't rise too fast.
• Stops harmful bacteria from growing in the dough.
• Makes the dough stronger and easier to handle.
• Tightens gluten, helping the dough hold gas better and rise evenly.
• Improves the texture and crumb of the baked bread.
• It holds moisture, it helps keep the bread soft and fresh longer.
• Usually used in small amounts (1–2% of the flour weight).
• If too little salt is added, yeast works too fast, uses up more sugar, and the bread crust becomes pale instead of golden brown.

4. Yeast

• The main function of yeast is to make the dough light and spongy and to give bread its pleasant aroma and flavour.
• The yeast used in bread making is **Saccharomyces cerevisiae**.
• Yeast uses the fermentable sugars present in the dough during the fermentation process.
• Through enzyme action, yeast produces **carbon dioxide gas and alcohol**.
• The carbon dioxide gas gets trapped in the gluten network, causing the dough to rise and become soft and airy.
• During fermentation, small amounts of alcohol and acids are also produced.
o These soften (mellow) the gluten, making the dough easy to stretch.
o The acids contribute to the characteristic flavour of the bread.
• Yeast is available in two main forms:
o **Compressed yeast (fresh yeast)**
o **Dried yeast (active dry yeast or instant yeast)**

20. Explain the types and processing of lozenges.

Lozenges – Types and Processing

What are Lozenges?

- Lozenges are **small candies** that usually contain medicine.

- They are **sucked and dissolved slowly** in the mouth.
- They are prepared by **cold mixing** of sugar, gum arabic/gelatine, stearine, and flavouring materials.

Types of Lozenges

There are **two main types**:

1. Plain Lozenges

- Made by **cutting pieces** from a sheet of dough.
- These pieces are then **dried in air**.
- No special shaping is needed.

2. Cut Lozenges

- Made from mixtures that are **rolled or extruded into sheets**.
- These sheets are then **cut into required shapes and sizes**.
- After cutting, they are **hardened by heating**.

Processing / How Lozenges Are Made

1. Mixing

- Ingredients like sugar, gum arabic/gelatine, stearine, and flavouring are mixed well.
- Proper mixing gives the lozenges good **shape and brittleness**.

2. Shaping

- Dough is **rolled or extruded** into sheets.
- Then **cut into desired shapes**.

3. Adding Colour & Flavour

- Colouring and flavouring can be added:
  - During kneading, or
  - During the hardening stage.

4. Drying

- The drying temperature should **not exceed 105°F**.
- Lozenges should not be kept longer than necessary in the drying stage.

21. Discuss how fat bloom occurs in chocolate.

FAT BLOOM

Fat bloom is the whitish or grayish layer that appears on the surface of chocolate or fat-based confectionery.
It happens when the fat inside the chocolate melts and moves to the surface, then solidifies in the wrong form.

Why Fat Bloom Happens

1. Chocolate contains cocoa butter, a type of fat.
2. When chocolate is stored in warm conditions, the fat melts.
3. When it cools again, the fat recrystallizes on the surface.
4. This forms a dull, whitish coat called fat bloom.

Key Features of Fat Bloom

• White/gray patches on chocolate
• Chocolate becomes soft, loses shine
• Texture becomes grainy and dull
• Taste remains safe, but quality reduces

Main Causes of Fat Bloom

1. Temperature changes (warm → cool → warm)
2. Improper tempering of chocolate
3. Fat migration from fillings (nut paste, cream, peanut butter)
4. Using different/low-quality fats
5. Long storage or poor packaging

How to Prevent Fat Bloom

1. Temper chocolate properly – so the fat sets correctly.
2. Store in cool place – Ideal: 18–20°C.
3. Avoid heat changes – no frequent shifting from hot to cold.
4. Avoid keeping in fridge – condensation causes bloom.
5. Use good-quality cocoa butter – poor-quality fats bloom faster.
6. Pack properly – use airtight, moisture-proof packaging.
7. Reduce oily fillings – excess fat in fillings can migrate out.

22. Define soft candy. Describe the processing of soft candy.

Definition of Soft Candy

Soft candy is a type of confectionery that is chewy, moist, and flexible. It has a smooth texture and does not break easily when bent. Examples include **toffees, caramels, and jelly candies**. Soft candies are made by heating sugar and other ingredients to a specific concentration and then cooling them to form a soft, pliable mass.

Processing of Soft Candy

1. Mixing of Ingredients

- Sugar, glucose, water, gelatin, fat, and other ingredients are mixed together to form a smooth, uniform paste.

2. Concentration

- The mixture is heated to remove excess water and make a thick syrup.
- This helps in achieving the correct sugar concentration and texture.

3. Cooling (Initial)

- The concentrated syrup is cooled slightly to form a “paste web” with tiny sugar crystals starting to develop.

4. Mixing & Homogenization

- The paste is stirred and pressed under pressure to make it smooth and consistent.

5. Shaping (Extrusion)

- The smooth paste is extruded into rods or other shapes for candies.

6. Cooling (Intermediate)

- The shaped candy is cooled to a manageable temperature for handling.

7. Final Mixing & Homogenization (Optional)

- Further mixing may be done to improve texture and uniformity.

8. Final Cooling

- Candy rods are cooled to room temperature to set properly.

9. Cutting & Packing

- The cooled candy is cut into pieces of suitable size and packed for sale.